AI-900 Revision Plan & Notes

Microsoft Azure AI Fundamentals – Complete exam-oriented revision notes

1. AI Fundamentals & Workload Foundations

- AI vs Machine Learning vs Deep Learning
- Training vs Inferencing
- ML problem types: Classification, Regression, Clustering
- AI-900 focuses on recognition, not calculations

2. Responsible AI Principles

- Fairness – avoid bias and discrimination
- Reliability & Safety – consistent outcomes
- Privacy & Security – protect user data
- Inclusiveness – accessible AI
- Transparency – explainable AI results
- Accountability – humans remain responsible

3. Azure AI Services vs Azure Machine Learning

- Azure AI Services: prebuilt models, no ML expertise required
- Azure Machine Learning: custom model training and management
- Training, tuning, evaluation → Azure Machine Learning
- Analysis, detection, extraction → Azure AI Services

4. AI Workloads and Azure Services

- Computer Vision → Azure AI Vision
- Natural Language Processing → Azure AI Language
- Speech → Azure AI Speech
- Document Intelligence → Azure AI Document Intelligence
- Generative AI → Azure OpenAI

5. Keyword to Service Mapping

- Image, photo, OCR → Azure AI Vision
- Sentiment, language, entities → Azure AI Language
- Speech, voice → Azure AI Speech

- Invoices, forms → Azure AI Document Intelligence
- Chatbots, text generation → Azure OpenAI
- Custom training → Azure Machine Learning

6. Training & ML Lifecycle (Conceptual)

- Data preparation and preprocessing
- Model training and evaluation
- Model deployment as endpoint
- Inferencing using the model
- Retraining with new data

7. Model Evaluation Metrics (Recognition Level)

- Classification: Accuracy, Precision, Recall, F1 score
- Regression: MAE, MSE, RMSE
- Clustering: Silhouette score
- Speech: Word Error Rate (WER)
- Generative AI: human evaluation

8. Exam Question Thinking Flow

- Identify the input type (image, text, audio, document)
- Map input to the correct AI workload
- Decide prebuilt vs custom model
- Choose the correct Azure service
- Eliminate incorrect options

These notes cover the full AI-900 syllabus at the correct exam depth and are intended for quick revision.